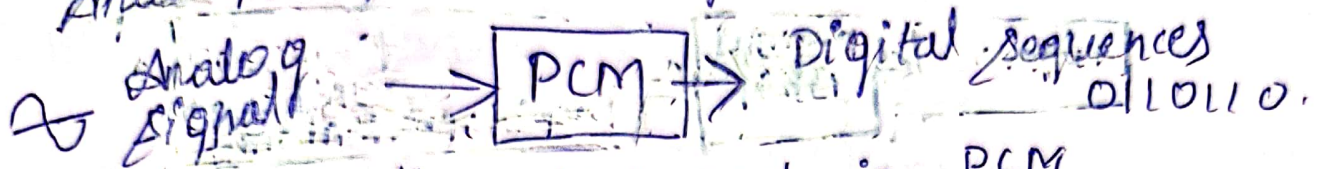


Module-5

PULSE CODE MODULATION:-

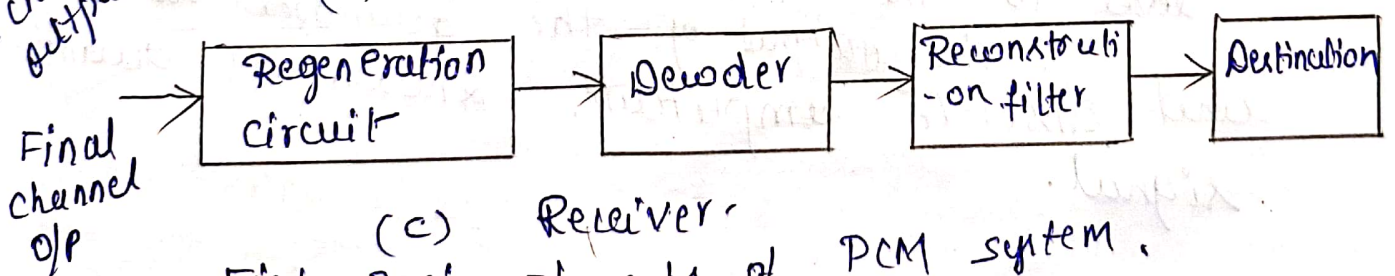
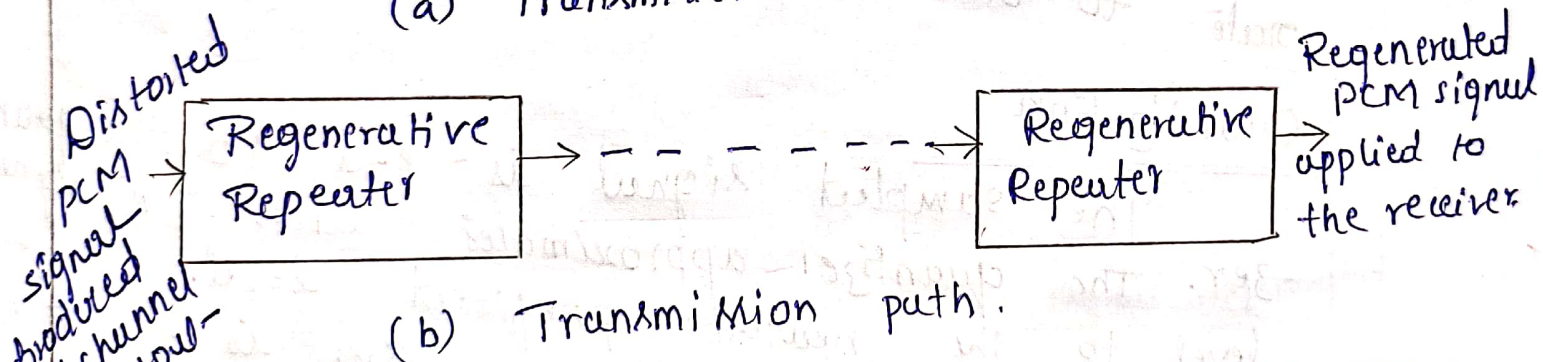
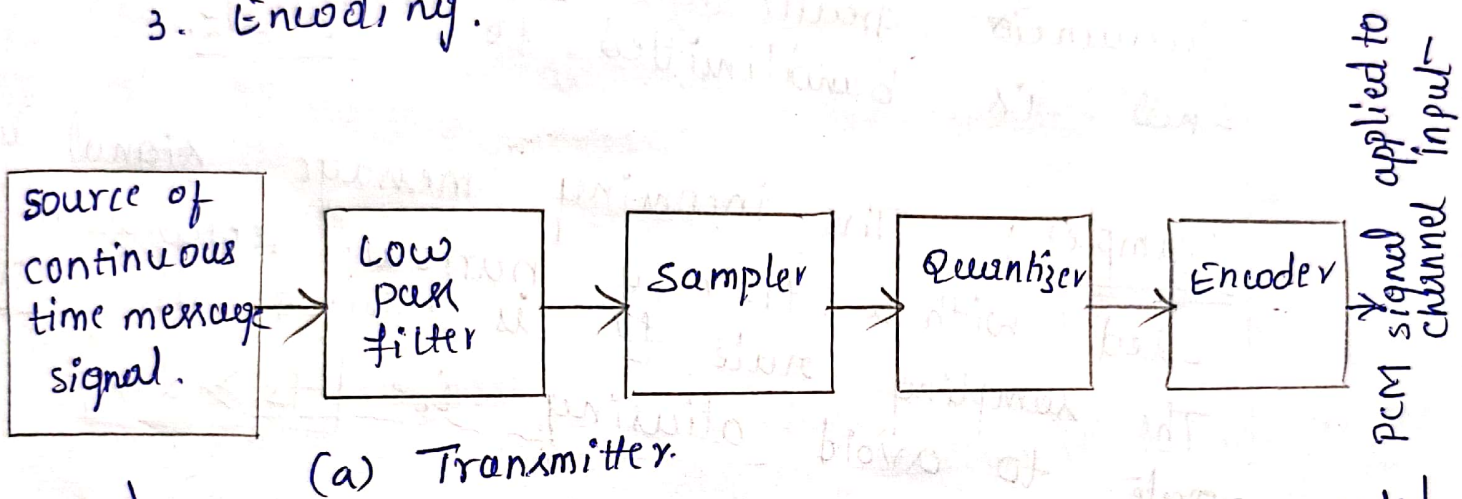
It is a digital pulse modulation.

Discrete sequence to
Analog signal to Digital signal conversion.



Basic operations performed in PCM

1. Sampling
2. Quantization
3. Encoding.



Fig! Basic Elements of PCM system.

* PCM converts analog information to digital sequences i.e. 1s and 0s.
* PCM is the most basic form of digital pulse modulation

* In PCM modulation, message signal is represented by a sequence of coded pulses, which is accomplished by representing the signal in discrete form in both time and amplitude.

The block diagram of a PCM system is shown in the above figure.

It consists of

- 1) Transmitter
- 2) Regenerative repeater
- 3) Receiver

PCM Transmitter: Low pass filter

In practice, the low pass filter (pre-alia filter) is used before sampler. in order to limit the frequencies greater than $\frac{W}{2}$ Hz. Hence message signal is bandlimited to $\frac{W}{2}$ Hz.

Sampler: The incoming message signal is sampled with a train of narrow rectangular pulses. The sampling rate f_s is selected above Nyquist rate to avoid aliasing i.e. $f_s \geq 2W$

Quantization

The sampled signal is fed to the quantizer. The quantizer approximates each I/P signal level to the nearest prefixed level. The output of the quantizer is discrete time and discrete amplitude. known as quantized signal.

Encoding: The quantized samples are then encoded in the encoder. The process of encoding involves allocating some digital code to each level. These coded levels are transmitted as a bitstream of data i.e. 0's and 1's

The encoder o/p consists of pulses depending on code combination.

Regenerative Repeater:

The PCM signal is reconstructed by means of a regenerative repeater, located at sufficiently closed spacing along the transmission path.

The regenerative networks are used at intermediate points, between transmitter and receiver in order to Boost up the pulse amplitude.

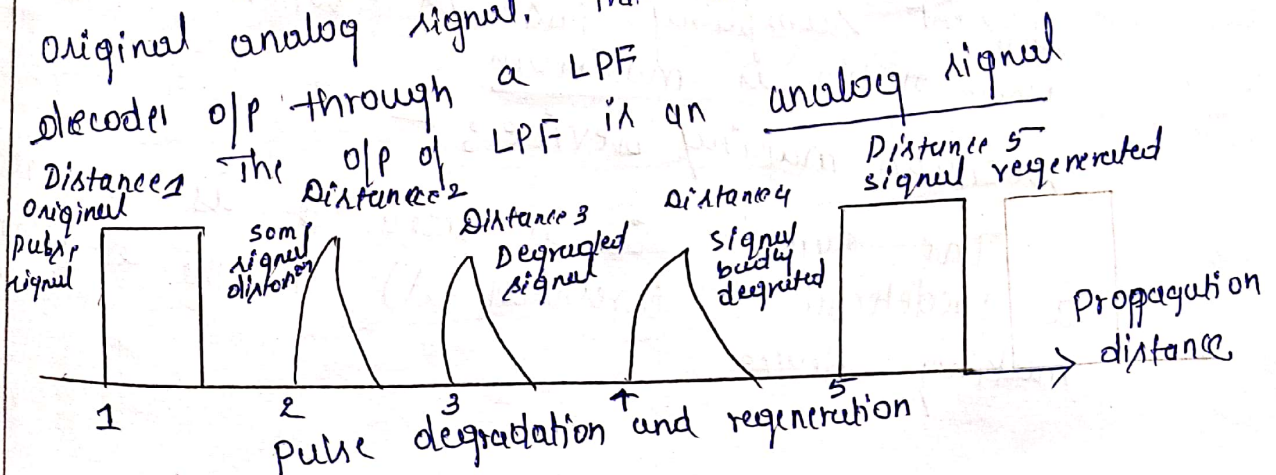
PCM Receiver:

Decoder: The 1st operation in the receiver is to generate the received pulses.

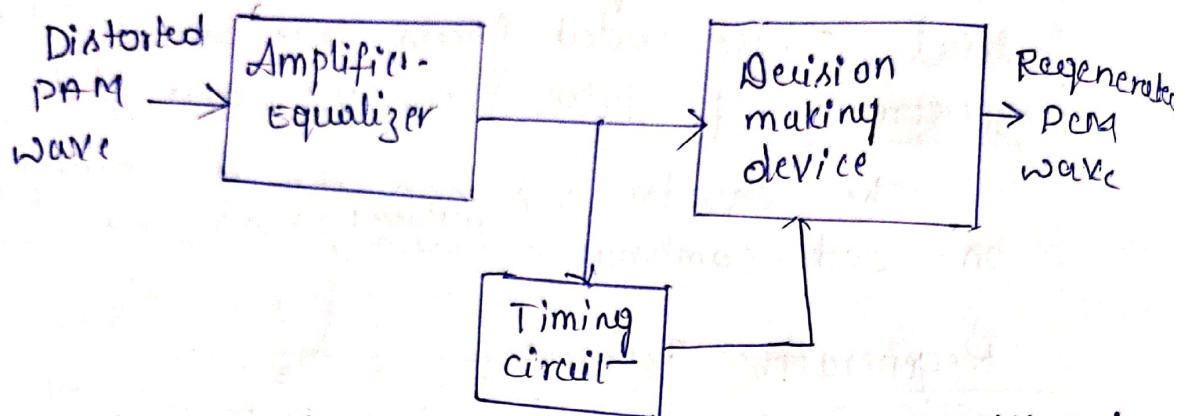
The decoder converts binary coded [1's and 0's] signal to a approximated pulses of discrete magnitude.

Reconstruction filter:

The final operation on the receiver is to recover the original analog signal. This is done by passing the decoder o/p through a LPF.



With a neat block diagram, explain the operation of a regenerative repeater



The regenerative repeater is used to remove noise from the incoming PCM data. Regenerative repeater is installed (placed) at regular interval along the channel through which PCM signal is transmitted.

The regenerative repeater perform three basic functions.

1) Equalizer (2) Timing (3) Decision making.
Equalizer: The equalizer shapes the received pulse so as to compensate for the effect of amplitude and phase distortion produced by the transmission channel.

Timing circuit:-

The timing circuit provides a periodic pulse train derived from the received pulses. The output of amplifier equalizer is sampled at different sampling instants where the signal to noise ratio is maximum.

Decision making device:-

The sample so is extracted is compared with a predetermined threshold (A) in the decision making device.

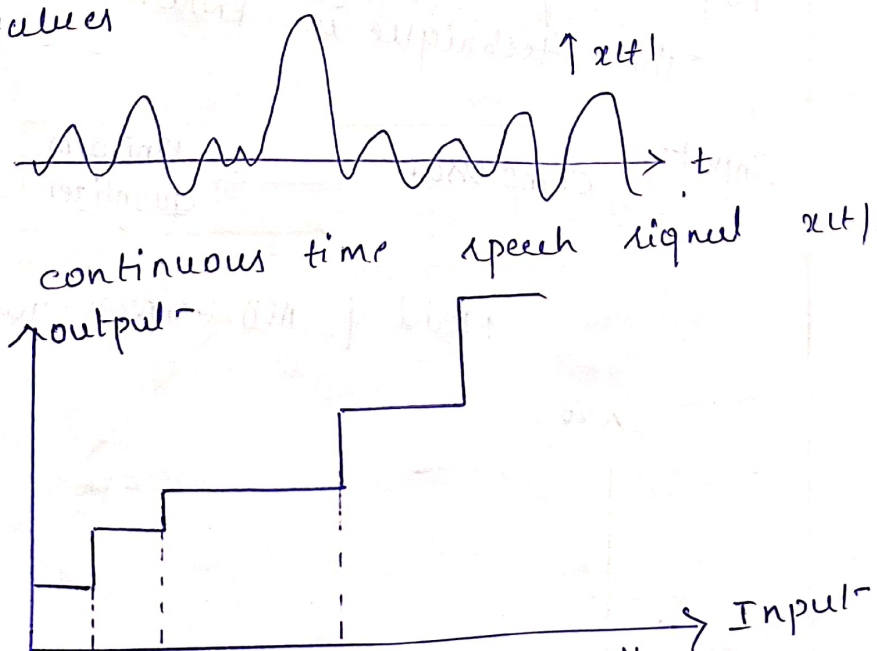
* During each bit interval, a decision is made, whether the received symbol is '1' or '0'.

* If the threshold is exceeded, a clean new pulse, representing symbol '1', is transmitted, otherwise a clean base line representing '0' is transmitted.

Explain the need for companding in PCM system. Describe compressor and expander characteristics.

* speech communication is very important in digital communication system. If uniform quantization is used, the step size will be constant.

* The system that uses equally spaced quantization levels, the quantization noise is same for all signal amplitudes. Hence small amplitude samples are more affected than the bigger sample values.



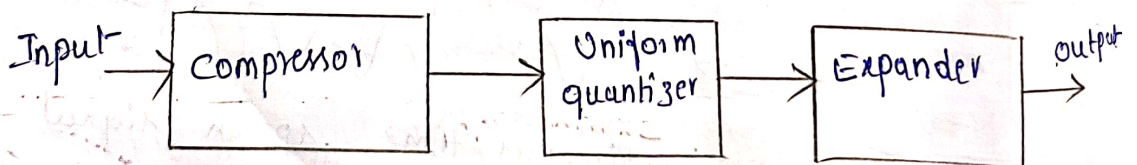
Transfer characteristic of a nonuniform quantizer.

∴ To keep signal to quantization noise ratio high, we must use a signal which is large in comparison with step size. This requirement is not satisfied when signal is small. i.e. we need smaller step size for low magnitude signal. samples need higher step size for higher magnitude signals.

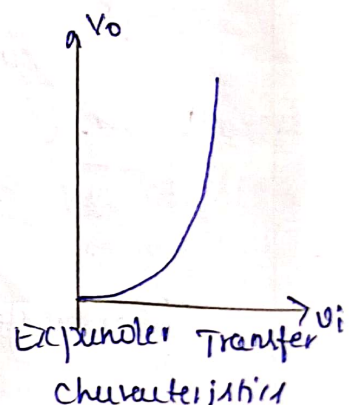
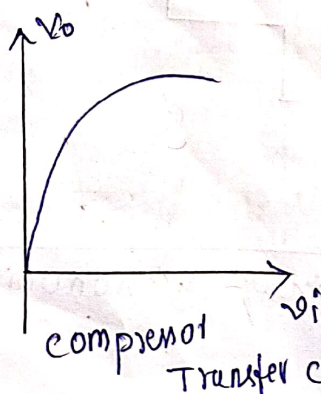
Changing step size according to signal magnitude is not preferable one. Instead, change the characteristics of signal such that lower amplitudes are amplified without changing the maximum value of the signal.

* Maintaining of constant SNR throughout the signal range is called "Robust quantization".

To achieve this, the signal is passed through a combination of compressor - Expander circuit respectively at the transmitter and receiver. This technique is known as COMPANDING.



Model of non-uniform quantizer



* To achieve robust quantization, the signal is passed through a network which has an I/p-o/p characteristic as shown in figure.

* Small amplitude signals are boosted up without altering the maximum amplitude of the signal, small amplitude signals range through more quantization levels.

* Any signal when passed through such a network gets compressed leading to signal distortion

* To remove this distortion, the signal is passed through an Inverse network at the receiver called as Expander.

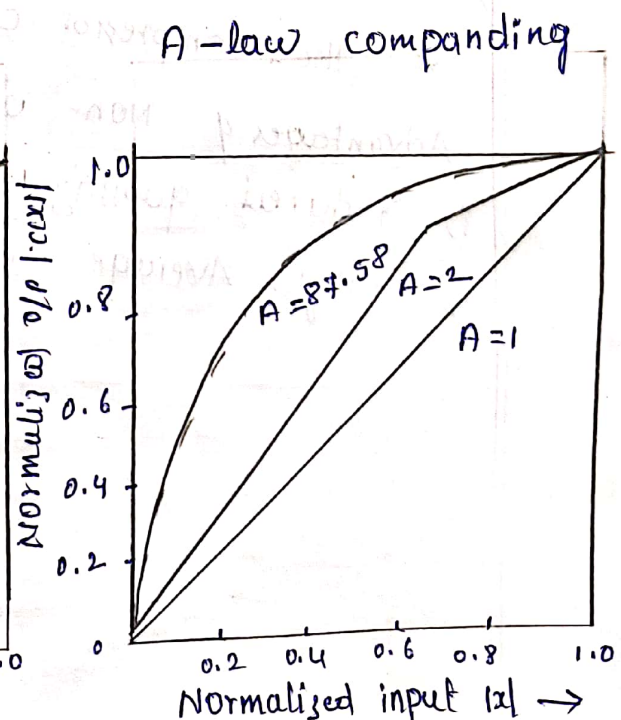
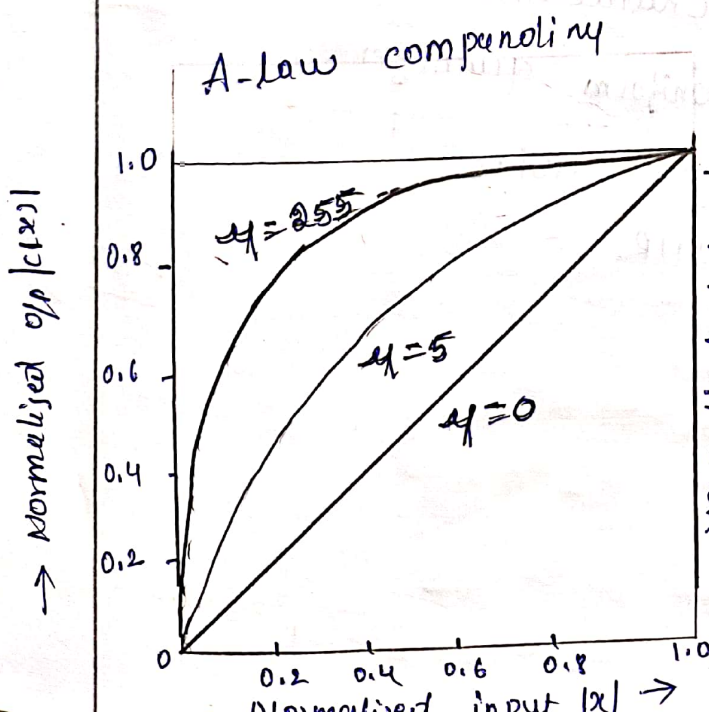
* The complete process of compression and expansion of the signal is called companding.

There are two types of companding

- 1) μ -law companding
- 2) A-law companding

Explain the following compression law.

- (a) μ -law (b) A-law



μ -law companding:

μ -law is defined by following equation

$$\frac{C(x)}{x_{\max}} = \frac{\ln(1 + \mu|x|/x_{\max})}{\ln(1 + \mu)} \quad 0 \leq \frac{|x|}{x_{\max}} \leq 1$$

μ -law companding:

- * It is used in United States, Canada and Japan
- * When $\mu=0$, compression corresponds to uniform quantization
- * In μ -law companding, the compression characteristics is continuous.

A-law companding:

$$\frac{C(x)}{x_{\max}} = \begin{cases} \frac{A|x|/x_{\max}}{1 + \ln A} & 0 \leq \frac{|x|}{x_{\max}} \leq \frac{1}{A} \\ \frac{1 + \ln(A|x|/x_{\max})}{1 + \ln A} & \frac{1}{A} \leq \frac{|x|}{x_{\max}} \leq 1 \end{cases}$$

- * It is used in India and Europe.
- * For digital telephony $A = \underline{87.6}$ is the standard value used in Europe
- * The compressor characteristics is piece-wise

Advantages of Non-uniform quantizer

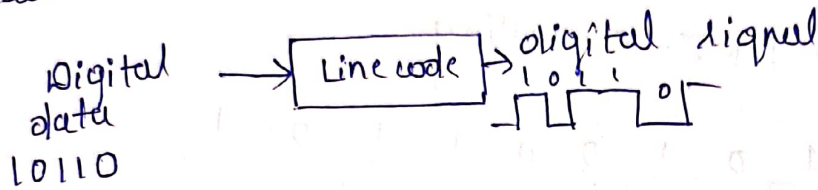
- 1) Reduced quantization noise
- 2) High Average SNR

Line codes [Discrete PAM signals].

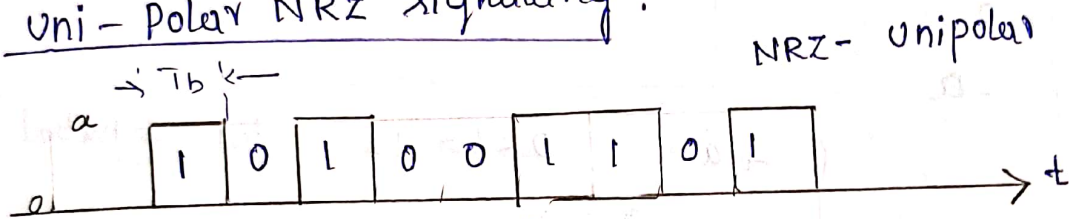
Also called as waveform coding

Line coding:

Digital data i.e. binary digits [1's and 0's] are represented by different waveforms for the purpose of transmission over the channel. This process is called line coding or transmission coding.



Uni-Polar NRZ signalling:

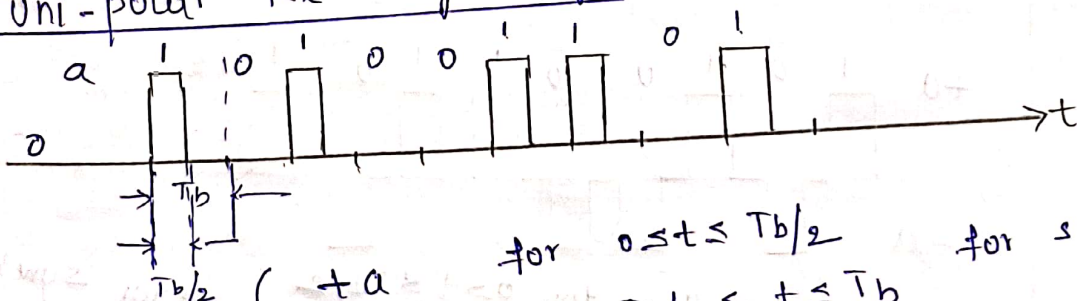


NRZ - Non-Return to Zero.

$$s(t) = \begin{cases} +a & \text{for } 0 \leq t \leq T_b \text{ for symbol 1} \\ 0 & \text{for } 0 \leq t \leq T_b \text{ for symbol 0} \end{cases}$$

When the pulse occupies the full duration of a symbol, the unipolar format is said to be of unipolar non-return to zero (NRZ) type

Uni-polar RZ-signalling:



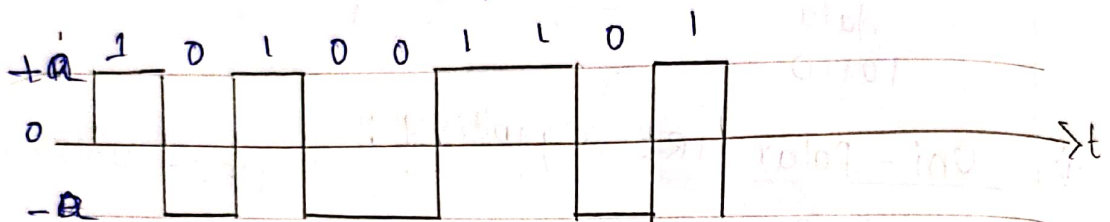
$$s(t) = \begin{cases} +a & \text{for } 0 \leq t \leq T_b/2 \text{ for symbol 1} \\ 0 & \text{for } T_b/2 \leq t \leq T_b \text{ for symbol 1} \\ 0 & \text{for } 0 \leq t \leq T_b \text{ for symbol 0} \end{cases}$$

When the pulse occupies the one-half of the symbol duration, it is said to be of non-return-to-zero (NRZ) type

Polar Format

In polar format, a positive pulse is transmitted for symbol 1, and a negative pulse for symbol 0.

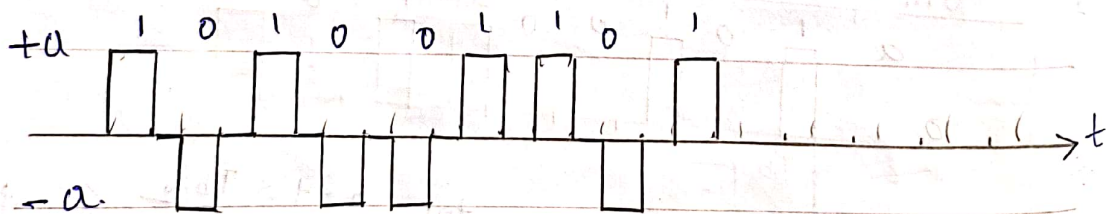
Polar NRZ-signalling



$$S(t) = \begin{cases} +a & 0 \leq t \leq T_b \text{ for symbol 1} \\ -a & 0 \leq t \leq T_b \text{ for symbol 0} \end{cases}$$

When the pulse occupies the full duration of a symbol, the unipolar format is said to be of Polar non-return-to-zero (NRZ) type

Polar RZ-signalling



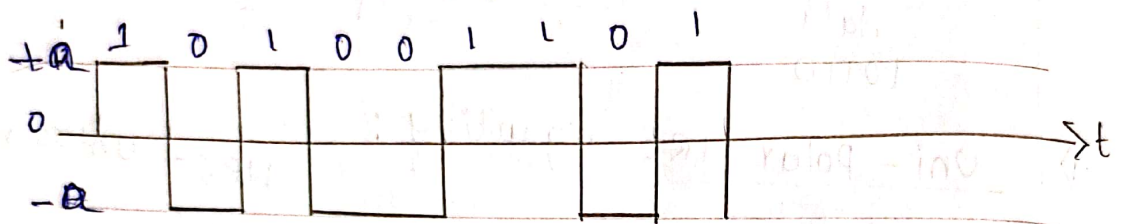
$$S(t) = \begin{cases} +a & \text{for } 0 \leq t \leq T_b/2 \rightarrow \text{for symbol 1} \\ 0 & \text{for } T_b/2 \leq t \leq T_b \\ -a & \text{for } 0 \leq t \leq T_b/2 \rightarrow \text{for symbol 0} \\ 0 & \text{for } T_b/2 \leq t \leq T_b \end{cases}$$

When the pulse occupies the one-half of the symbol duration, it is said to be of non-return-to-zero (NRZ) type

Polar Format

In polar format, a positive pulse is transmitted for symbol 1, and a negative pulse for symbol 0.

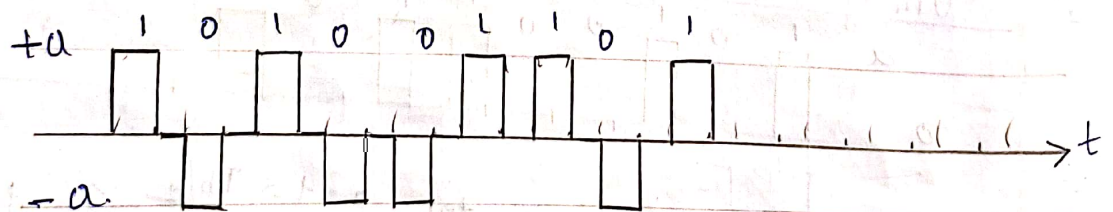
Polar NRZ-signalling



$$s(t) = \begin{cases} +a & 0 \leq t \leq T_b \text{ for symbol 1} \\ -a & 0 \leq t \leq T_b \text{ for symbol 0} \end{cases}$$

When the pulse occupies the full duration of a symbol, the unipolar format is said to be of polar non-return-to-zero (NRZ) type

Polar RZ-signalling

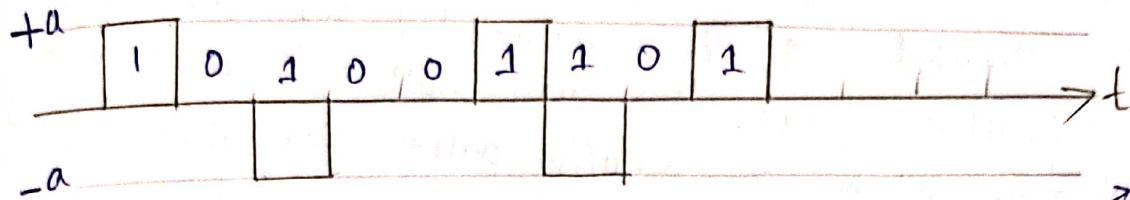


$$s(t) = \begin{cases} +a & \text{for } 0 \leq t \leq T_b/2 \rightarrow \text{for symbol 1} \\ 0 & \text{for } T_b/2 \leq t \leq T_b \\ -a & \text{for } 0 \leq t \leq T_b/2 \rightarrow \text{for symbol 0} \\ 0 & \text{for } T_b/2 \leq t \leq T_b \end{cases}$$

Bipolar Format or Pseudo ternary signalling

In bipolar format, positive and negative pulses are used alternately for transmission of 1's and no pulse for the transmission of 0's

Bipolar NRZ- signalling

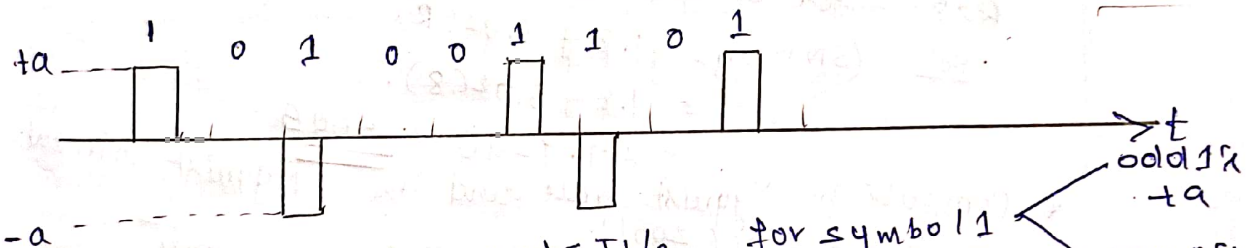


$$s(t) = \begin{cases} \pm a & \text{for } 0 \leq t \leq T_b \text{ for symbol 1} \\ 0 & \text{for } 0 \leq t \leq T_b \text{ for symbol 0.} \end{cases}$$

odd 1's $\rightarrow +a$
even 1's $\rightarrow -a$

When the pulse occupies the full duration of a symbol, the unipolar format is said to be of BIPOLAR non-return to zero (NRZ) type.

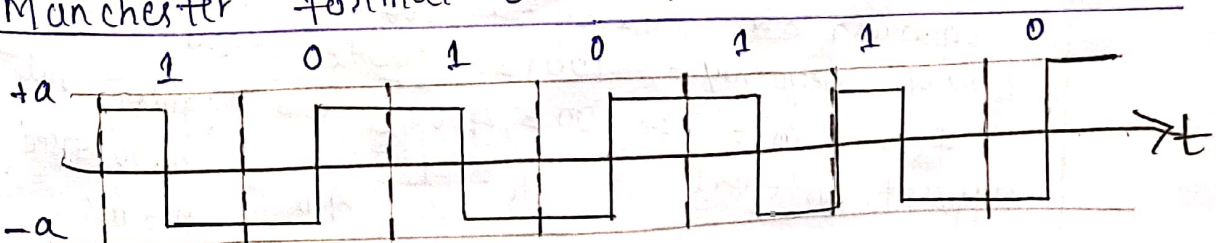
Bipolar RZ signalling:



$$s(t) = \begin{cases} \pm a & \text{for } 0 \leq t \leq T_b/2 \text{ for symbol 1} \\ 0 & \text{for } T_b/2 \leq t \leq T_b. \\ 0 & \text{for } 0 \leq t \leq T_b \text{ for symbol 0.} \end{cases}$$

odd 1's $\rightarrow +a$
even 1's $\rightarrow -a$

Manchester format or Biphasic or split phase



$$S(t) = \begin{cases} +a & \text{for } 0 \leq t \leq T_b/2 & \text{for symbol 1} \\ -a & \text{for } T_b/2 \leq t \leq T_b \\ -a & \text{for } 0 \leq t \leq T_b/2 & \text{for symbol 0} \\ +a & \text{for } T_b/2 \leq t \leq T_b \end{cases}$$

Symbol 1 is represented by transmitting a positive pulse for one-half of the symbol duration followed by a negative pulse for the remaining half of the symbol duration.

Symbol 0 is represented by transmitting a negative pulse for one-half of the symbol duration, followed by a positive pulse for the remaining half of the symbol duration.

Manchester waveform has no dc component.

Module - 4 problems

* Compute the output SNR in dB for number of bits/sample $R=8$ for a quantizer.

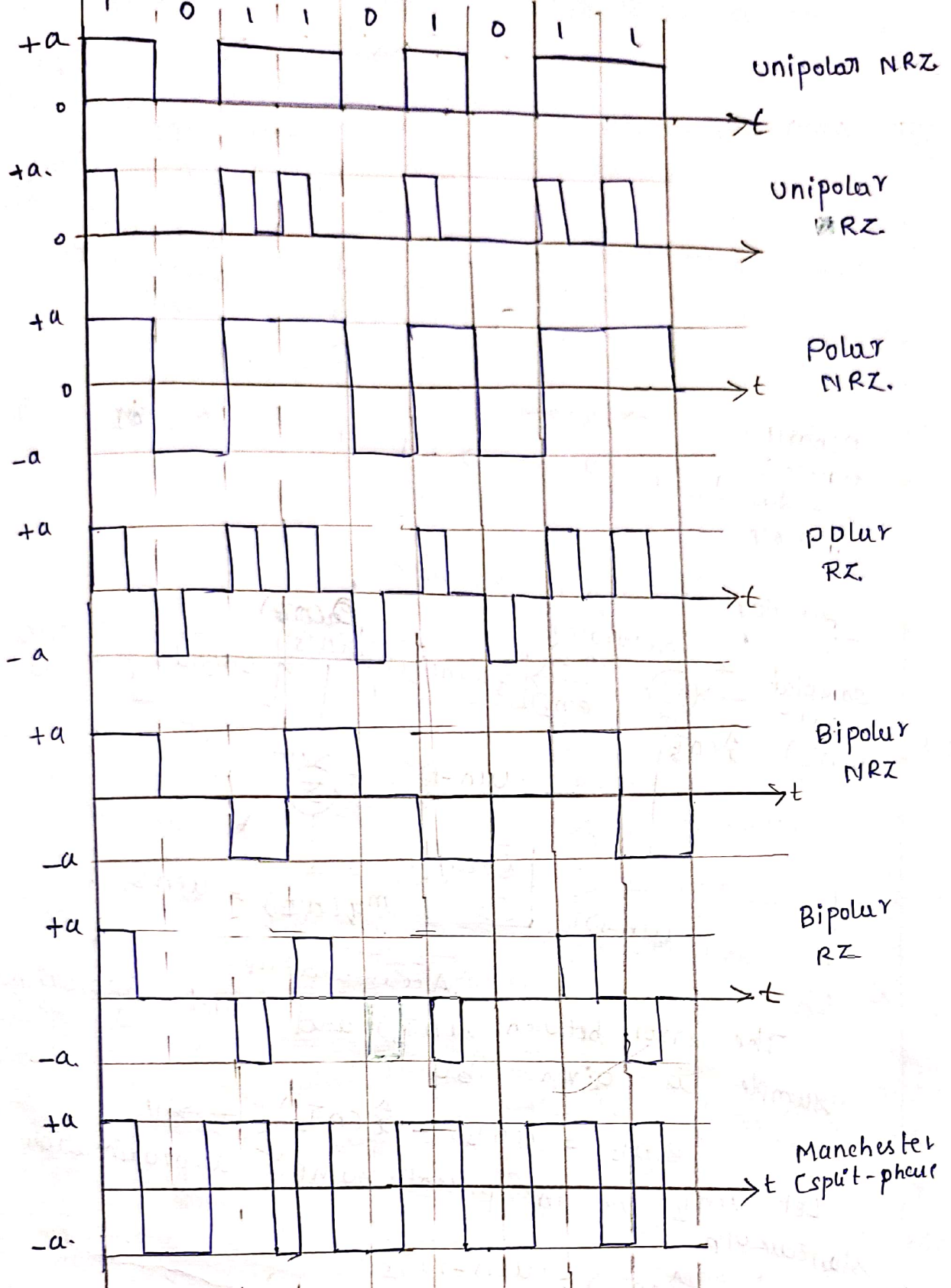
$$\text{Sol } (SNR)_{dB} = 1.8 + 6.02(R) \\ = 1.8 + 6.02(8) \\ = 49.92 \text{ dB} = \underline{50 \text{ dB}}$$

* Compute the Nyquist rate and the Nyquist interval for signal $g(t) = \text{sinc}(200t)$

$$\omega = 200 \quad \text{Nyquist sampling rate} \\ f_m = \omega = \frac{200}{2} = 100 \quad f_s = 2\omega = 200 \text{ Hz} \\ \text{Nyquist Interval} \quad T_s = \frac{1}{f_s} = \frac{1}{200} = 0.005 \text{ s} = \underline{5 \text{ ms}}$$

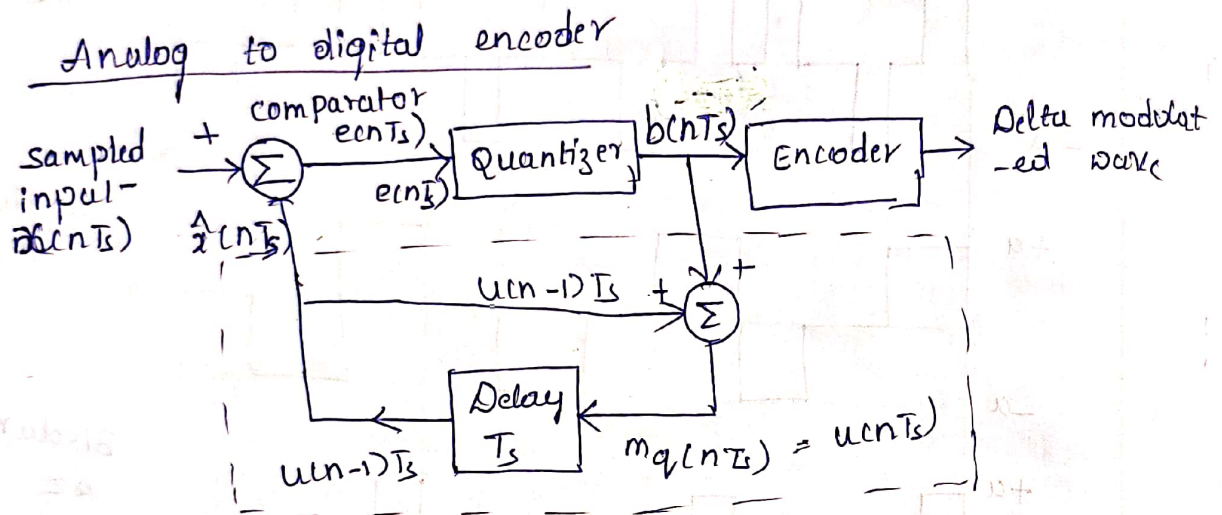
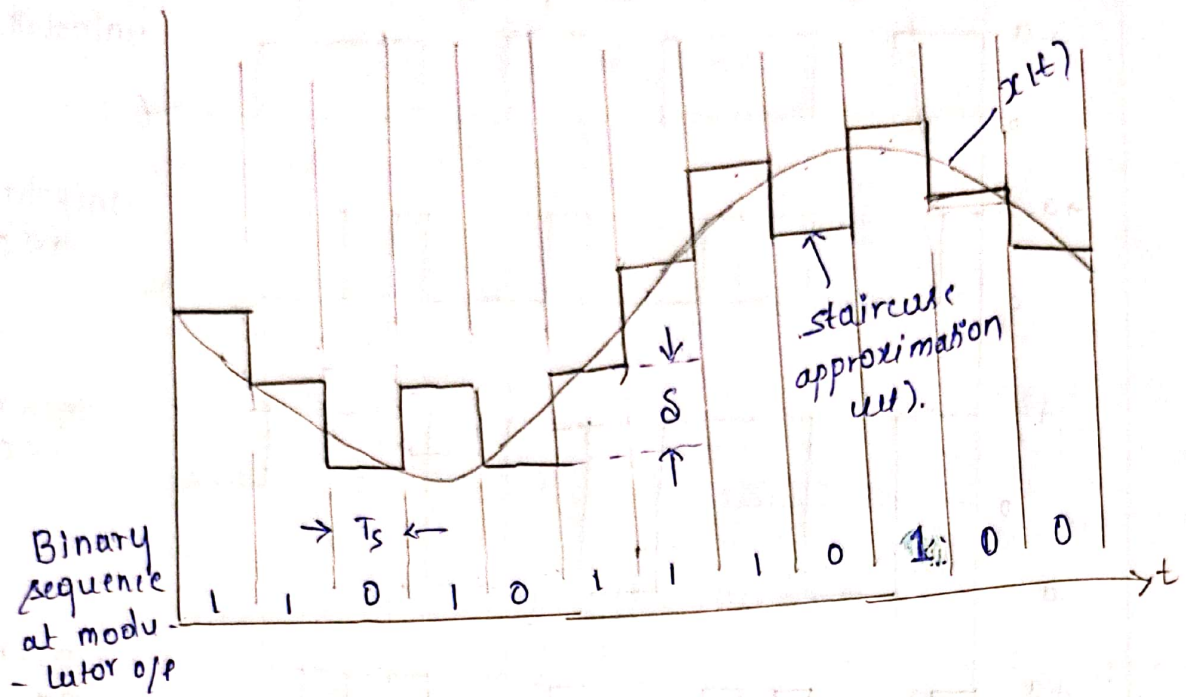
* Compute the Nyquist rate and Nyquist interval for signal $g(t) = \text{sinc}^2(200t)$

$$\text{Double frequency} = 100 \times 2 = 200 \text{ Hz} = f_m = \omega \\ f_s = 2f_m = 2 \times 200 = 400 \text{ Hz} \rightarrow \text{Nyquist rate} \\ \text{Nyquist interval} \quad T_s = \frac{1}{f_s} = \frac{1}{400} = 0.0025 \text{ s} = \underline{2.5 \text{ ms}}$$



Delta modulation:

Delta modulation transmits only one bit per sample i.e. the present value is compared with the previous sample value and the indication, whether the amplitude is increased or decreased is sent.



The error between $x(nT_s)$ and last approximated sample is given by

$$e(nT_s) = x(nT_s) - \hat{x}(nT_s) \rightarrow (1)$$

Let $u(nT_s)$ be the present sample approximation of staircase o/p.

$$\hat{x}(nT_s) = u(n-1)T_s$$

$$\hat{x}(nT_s) = u(nT_s - T_s) \rightarrow (2)$$

Substitute equation (2) in equation (1), we get

$$e(nT_s) = x(nT_s) - u(nT_s - T_s) \rightarrow (3)$$

The binary quantity $b(nT_s)$ is the algebraic sign of the error $e(nT_s)$, except for the scaling factor Δ . i.e. $b(nT_s) = \Delta \text{sgn}[e(nT_s)] \rightarrow (4)$

$b(nT_s)$ depends on the sign of error $e(nT_s)$
the sign of step-size Δ
will be decided.

i.e. $b(nT_s) = +\Delta$ if $x(nT_s) > \hat{x}(nT_s)$

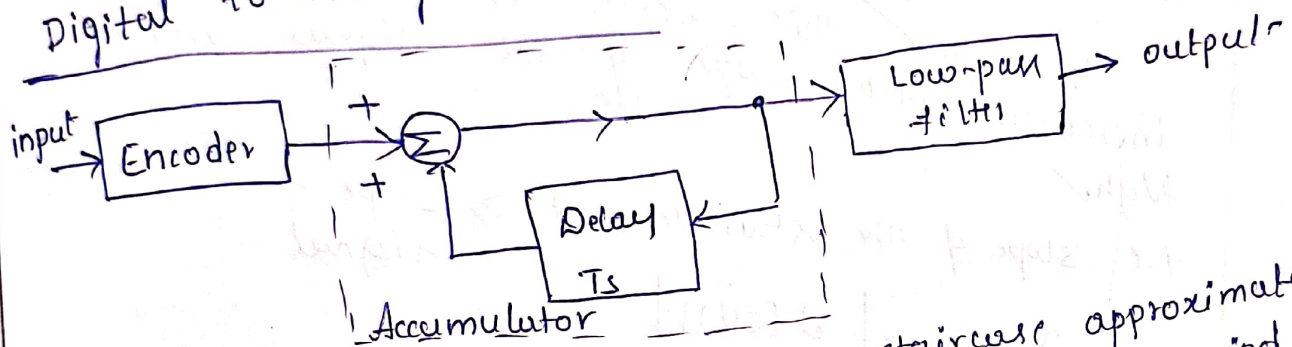
$b(nT_s) = -\Delta$ if $x(nT_s) \leq \hat{x}(nT_s)$

If $b(nT_s) = +\Delta$ then binary '1' is transmitted

If $b(nT_s) = -\Delta$ then binary '0' is transmitted

$\therefore u(nT_s) = u(nT_s - T_s) + b(nT_s)$

Digital to analog decoder.



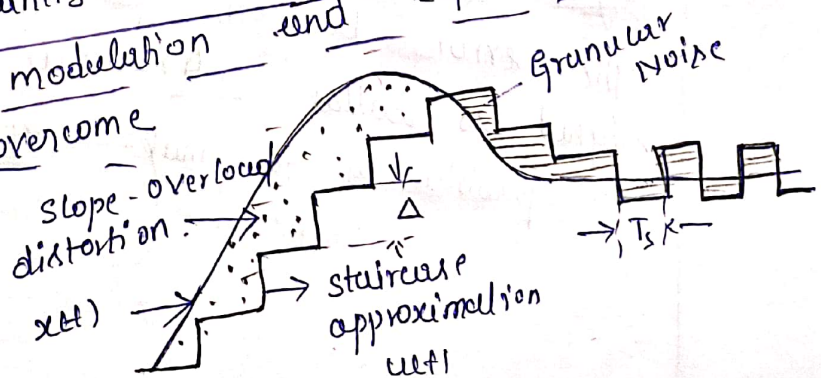
* The accumulator generates the staircase approximated signal o/p and it is delayed by one sampling period ' T_s '. It is then added to the i/p signal.

* If I/p is binary '1' then it adds $+\Delta$ step to the previous o/p.

* If I/p is binary '0' then one step ' Δ ' is subtracted from the delayed signal.

* The LPF is used to remove step variation and to get smooth reconstructed message signal $\hat{x}(t)$.

Illustrate the quantization error, such as granular noise error could be overcome



Slope overload distortion

Slope overload distortion arises because of the large dynamic range of the input signal.

In fig it can be seen, the rate of rise of i/p signal $x(t)$ is so high that the staircase signal cannot approximate it, the step size Δ become too small for staircase signal $\hat{x}(t)$ to follow the steep segment of $x(t)$. Thus large error between the staircase approximated signal and the original i/p signal $x(t)$. This error is called slope overload distortion.

To reduce this error, the step size should be increased when slope of the signal $x(t)$ is high.

i.e slope of the staircase $\hat{x}(t) \geq$ slope of the message signal

$$\frac{\Delta}{T_s} \geq \max \left| \frac{d[x(t)]}{dt} \right|$$

Granular noise

This noise occurs when the step size is too large compared to small variations in the input signal, i.e for very small variations in input signal, the staircase signal is changed by large amount because of large step size Δ .

In the figure signal is almost flat, the staircase signal $\hat{x}(t)$ keeps on oscillating by $\pm \Delta$ around the signal.

The error between the input and approximated signal is called Granular Noise. The solution of this problem is to make step size small.

Time Division Multiplexing (TDM)

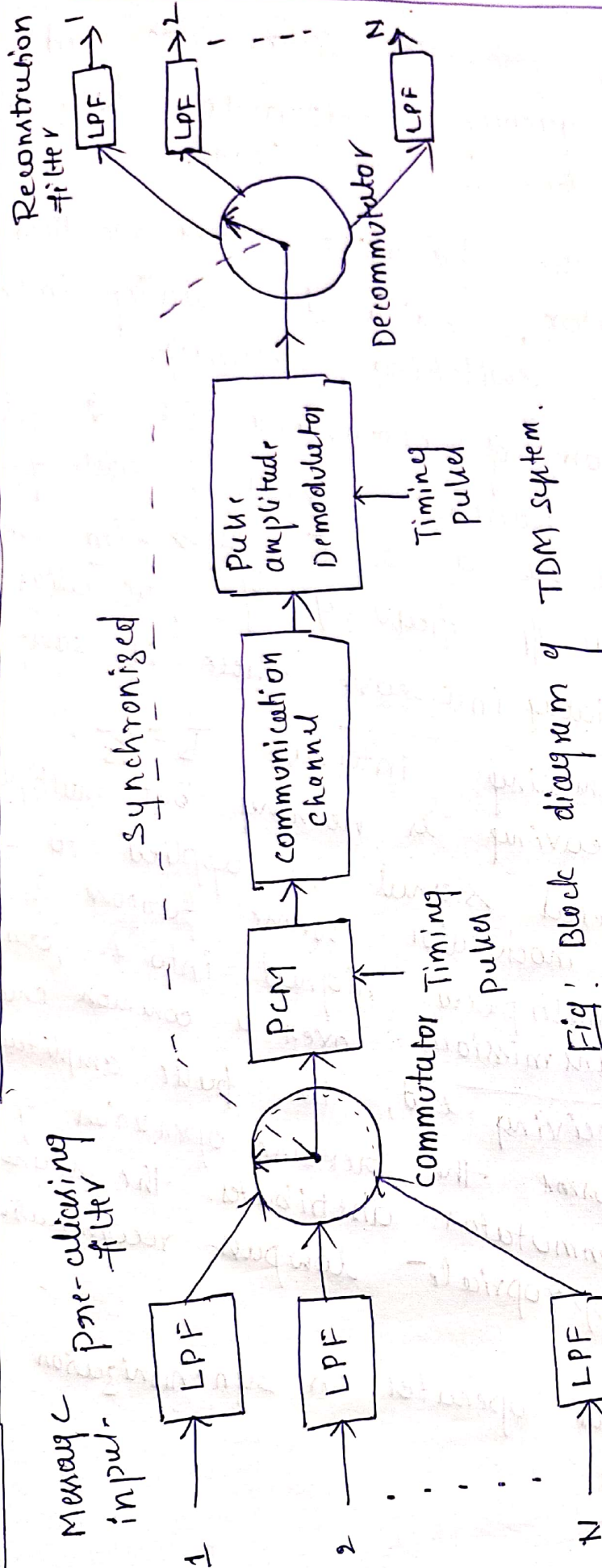


Fig. Block diagram of TDM system.

In TDM, different time intervals are allocated for different message signals, so that a common channel is utilized for transmission of these signals without any interference.

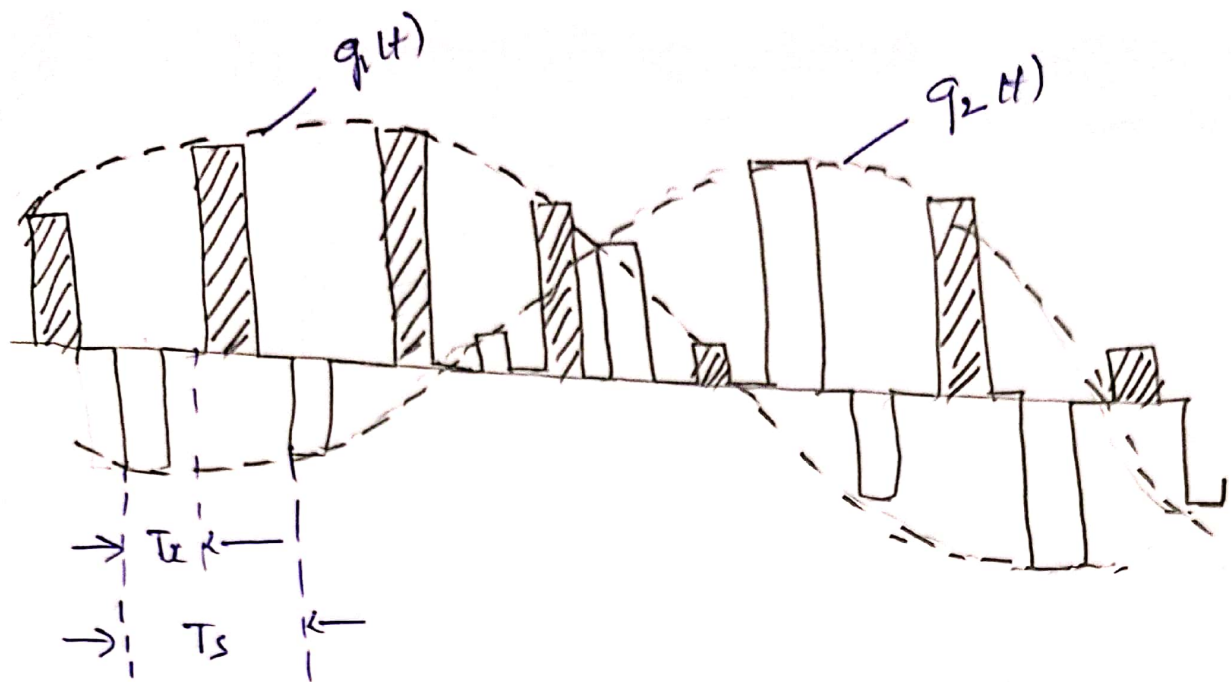
The concept of TDM is illustrated in the block diagram.

- * The low pass pre-alias filters are used to remove high frequency components which may be present in the message signal.
- * The o/p of the pre-alias filters are then fed to a Commutator, which is usually implemented using electronic switching circuitry.
- * The function of commutator is 2 fold:
 - 1) To take a narrow samples of each of the N i/p signals at a rate $f_s \gg 2f_m$, where ' f_m ' is the cut-off frequency of pre-alias filter.
 - 2) To sequentially interleave these ' N ' samples inside a sampling interval $T_s = \frac{1}{f_s}$.

This interleaving is nothing but multiplexing.

- * The multiplexed signal is applied to a pulse amplitude modulator whose purpose is to transform the multiplexed signal into a form suitable for transmission over a common channel.

- * At the receiving end, the pulse amplitude demodulator performs the reverse operation of PAM and the decommutator distributes the signals to the appropriate low pass reconstruction filters.
- * The decommutator operates in synchronization with the commutator.



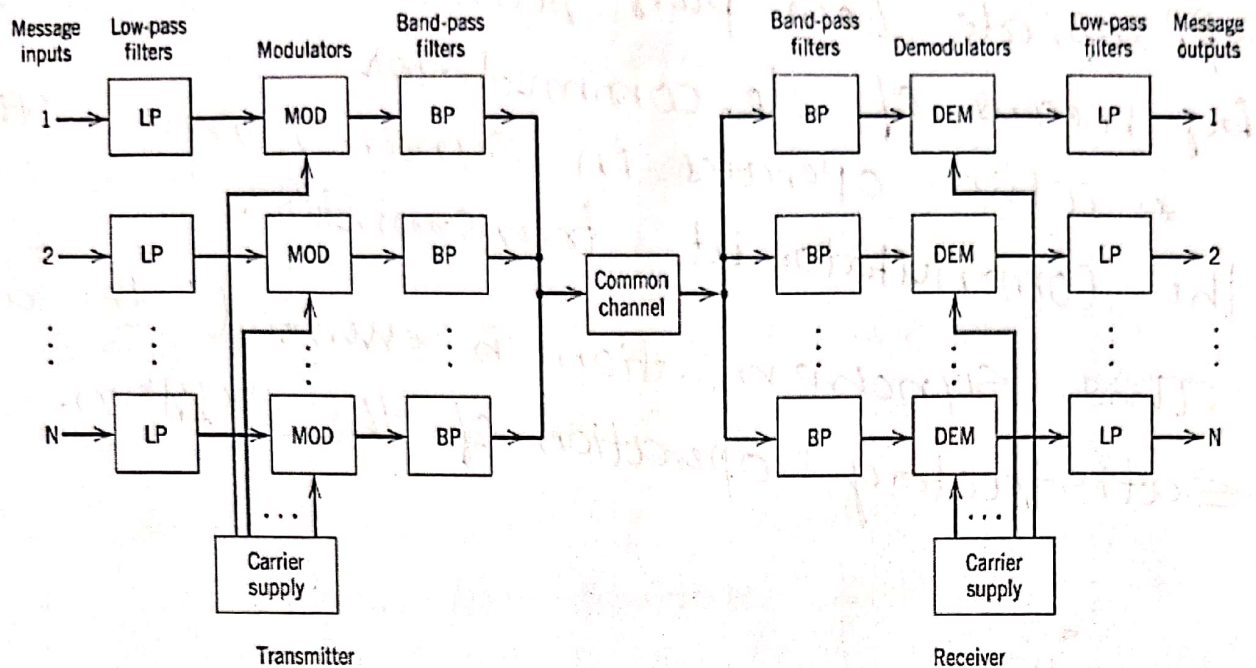
$T_s \rightarrow$ denotes the sampling period

$T_x \rightarrow$ denote the time spacing between adjacent-sampler in the TDM signals

i.e. $T_x = \frac{T_s}{N}$ as shown in figure

$N \rightarrow$ Total number of message signals to be multiplexed.

Frequency Division Multiplexing (FDM)



Block diagram of FDM system.

Frequency Division multiplexing (FDM) is a multiplexing technique that divides the available bandwidth into ~~sub~~ multiple sub-bands each of which can carry a signal without overlap.

The frequency bands are separated from one another by strips of unused frequencies called the guard band, to prevent the overlapping of signals.

In FDM system, the input message signals assumed to be lowpass type are passed through the LP low pass filter. to remove the high frequency components present in the LP signal.

The filtered message signals are then modulated with necessary carrier frequencies with help of modulator. The most commonly used method of modulation in FDM is single sideband modulation.

Then applied to the band pass filter to restrict the band of each modulated wave to its prescribed range.

The outputs of band-pass filters are combined in parallel which form the input to the common channel.

At the receiving end, band pass filters connect to the common channel separate the different signals then applied to the demodulator to obtain the original message signals followed by lowpass filters.